

MediKiosk / HAMS - Technology Stack

Hospital & Healthcare Management System (ABDM, FHIR R4 & AYUSH Ready)
Generated for: Smart India Hackathon & Healthcare Innovation

1. Frontend Technologies (Client-Side)

- **Core Framework:** React 18 (Component-driven SPA architecture)
- **Build & Bundler:** Vite 5 (Lightning-fast HMR and ESM bundling)
- **Language:** TypeScript (Strict static typing & interfaces)
- **Styling System:** Tailwind CSS, PostCSS & Autoprefixer
- **Animations & UI:** Framer Motion & Canvas Confetti
- **Icons & Visuals:** Lucide React (Vector icons)
- **Analytics & Charts:** Recharts (Vital signs, triage and hospital analytics)
- **QR & Document Export:** QRCode.react, jsPDF (PDF reports & token generation)
- **Speech & Voice Engine:** Web Speech API (STT & TTS across 11 Indian Languages)

2. Backend Technologies (Server-Side)

- **Runtime Environment:** Node.js (v18+ / v20+ / v24+)
- **Web Framework:** Express.js (RESTful APIs, routing & middlewares)
- **Real-Time Communication:** Socket.io (Live emergency triage alerts & sync)
- **Security & Headers:** Helmet, CORS, Express-Rate-Limit
- **Authentication & Tokens:** JSON Web Tokens (JWT), BcryptJS (Password hashing)
- **File & Document Handling:** Multer (Prescription & lab report uploads)
- **Spreadsheet Processing:** SheetJS / XLSX (Patient & hospital excel data export)
- **HTTP Logger:** Morgan (Dev logging and API inspection)

3. Database & Data Storage Solutions

- **Primary Cloud Database:** Supabase (PostgreSQL 15 with Row Level Security)
- **NoSQL Support:** MongoDB (Mongoose ODM connectivity)
- **Resilient Offline Store:** In-Memory Cache & JSON Local Store (Zero-config offline mode)
- **Relational Entities:** 17 Tables (Patients, Visited Doctors, Triage, AYUSH, Audits)

4. Clinical AI Engines & Healthcare Standards

- **Clinical Questioning:** Adaptive SOCRATES Question Tree (16 Clinical Branches)
- **Emergency Triage:** Red-Flag & Yellow-Flag Screening (Code Red routing)
- **AYUSH Framework:** Dashavidha & Ashtavidha Pariksha + Prakriti Assessment
- **OCR & Drug Safety:** Prescription OCR Scanner & Drug-Drug Interaction Matrix
- **National Standards:** ABDM (ABHA M1/M2/M3 Sandbox) & DPDP Act 2023 Consent
- **Interoperability:** HL7 FHIR R4 Compliant Document Bundle Exporter